

UMAID SHAHZAD MUGHAL

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SUMMARY

Motivated junior Full Stack Developer with hands-on experience building production-grade MERN and PERN (MySQL) applications. Proficient in designing RESTful APIs, implementing JWT-based authentication, and structuring scalable backend architectures. Driven by a systems-thinking mindset and a continuous interest in backend engineering, clean code practices, and emerging technologies. Seeking an internship or junior role to apply real-world project experience to impactful engineering challenges.

EDUCATION

University of Engineering and Technology Lahore
B.S. in Computer Science | CGPA: 3.219 (4 Semesters)

Pakistan
Expected Graduation: June 2028

EXPERIENCE

NCAI UET Lahore | Full Stack Developer Intern | 28 June 2026 – 4 September 2026

- Engineered and optimized backend services using Node.js and Express.js, focusing on scalable MVC architecture and RESTful API design.
- Structured database schemas and managed data workflows utilizing MongoDB, ensuring efficient query performance and seamless integration with React.js frontends.
- Integrated external APIs and implemented secure user authentication workflows using JWT to protect sensitive endpoints and application data.
- Troubleshooted system bottlenecks, debugged applications, and maintained version control utilizing Git and GitHub in a collaborative environment.

PROJECTS

MH Concepts — Full Stack Interior Decor Lead Generation Platform - [Live Demo](#)

- Engineered a dynamic product showroom with real-time category filtering and keyword search, enabling a mobile-first browsing experience for business clients.
- Built a secure admin dashboard with full product CRUD, customer inquiry tracking, and live business analytics — all protected via JWT authentication and HTTP-only cookies.
- Integrated WhatsApp Cloud API and Nodemailer SMTP to deliver dual instant notifications to the admin on every new customer lead.
- Managed media assets through Cloudinary API for high-resolution image uploads; optimized MongoDB Atlas indexing for fast product retrieval at scale.

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, JWT, Cloudinary, WhatsApp Cloud API, Nodemailer, CORS

- CRM — Real-Time Customer Support & AI Fallback Platform - [Live Demo](#)

- Architected a scalable WebSocket (Socket.io) infrastructure using a Redis to enable instant, synchronized live chat between external leads and internal sales agents across multiple server instances.
- Integrated the Deepseek AI API to automatically intercept unassigned chats after a 30-second timeout, acting as a fallback chatbot to parse lead contact details, summarize conversations, and provide instant customer support.
- Engineered a global distributed state system using Redis to track real-time agent availability, routing incoming chats into a live queue that instantly syncs across all connected admin and agent consoles.
- Built a comprehensive Admin Dashboard using Recharts to visualize real-time CRM data, dynamically rendering formulas for lead sources, conversation statuses, and agent engagement metrics. **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Redis, Socket.io, Deepseek AI API, Tailwind CSS, Recharts

AI Based RAG Study Assistant

- Engineered a semantic RAG chat engine using gemini-embedding-001 and MongoDB Atlas Vector Search, converting PDF text into 768-dimensional coordinates queried via Cosine Similarity.
- Secured document-isolated workspaces using BSON pre-filtering inside the vector search pipeline; generated automated JSON study

companions (Summaries, MCQs, Flashcards) cached in MongoDB.
– Designed a synchronized presentation slide player utilizing browser-native Web Speech API to stream spoken narrator scripts in multiple languages with native accents, alongside cascading collection deletes.
Tech Stack: *React.js, Node.js, Express.js, MongoDB Atlas, MongoDB Atlas Vector Search, Gemini API, JWT, Web Speech API*

TECHNICAL SKILLS

Languages: JavaScript, C++, Python
Frontend: React.js, HTML5, CSS3, React Router DOM, Axios
Backend: Node.js, Express.js, REST APIs, JWT Authentication, MVC Architecture
Databases: MongoDB, MongoDB Atlas, MySQL, Raw SQL
Tools & Platforms: Git, GitHub, Postman, Docker, Antigravity , Claude , VS Code, MongoDB Compass, Cloudinary
Concepts: Backend Architecture, System Design Basics, Auth & Authorization, CRUD Operations, Relational DB Design

CERTIFICATIONS

Introduction to Front-End Development Jul 2025
Meta / Coursera | Verified: coursera.org/verify/LTCONZRIGLF9